



Cloud Infrastructure Analyst

QP Code: SSC/Q8304

Version: 4.0

NSQF Level: 6

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Qualification Pack

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SSC/Q8304: Cloud Infrastructure Analyst

Brief Job Description

Individuals in this job are responsible for designing, maintaining, evaluating, and testing architectures of cloud products and solutions. Cloud Architects also create enterprise policies and guidelines for cloud implementation and governance. They assess cloud vendors and third-party cloud service providers. They also assist in validating customer proposals and project plans for cloud implementation

Personal Attributes

Cloud Infrastructure Analysts must be able to communicate effectively, build relationships with others and collaborate with various stakeholders. He/she must have a zeal to learn new things.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [SSC/N8313: Provision, configure, test and deploy cloud infrastructure](#)
2. [SSC/N8314: Maintain compute, network and storage systems deployed on cloud and ensure backup and recovery](#)
3. [SSC/N8315: Automate cloud infrastructure management](#)
4. [SSC/N8316: Migrate legacy/ on-premise/cloud applications and data to new cloud platforms](#)
5. [SSC/N8317: Integrate cloud infrastructure with existing IT systems](#)
6. [DGT/VSQ/N0102: Employability Skills \(60 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	IT-ITeS
Sub-Sector	Future Skills
Occupation	Cloud Computing
Country	India
NSQF Level	6
Credits	20

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Aligned to NCO/ISCO/ISIC Code	NCO-2015/3513.0301
Minimum Educational Qualification & Experience	Completed 4 year UG program (PG/UG/Degree/Diploma in (Engg./Science)) OR Completed 3 year UG degree (PG/UG/Degree/Diploma in (Engg./Science)) with 1.5 years of experience in relevant field. Relevant Experience in job roles related to IT/Computer Science/Cloud Computing The relevant experience would include work, internship and apprenticeship after completion of relevant educational qualification. OR Previous relevant Qualification of NSQF Level (5) with 3 Years of experience in relevant field. Relevant Experience in job roles related to IT/Computer Science/Cloud Computing The relevant experience would include work, internship and apprenticeship after completion of relevant educational qualification.
Minimum Level of Education for Training in School	
Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	18/02/2028
NSQC Approval Date	18/02/2025
Version	4.0
Reference code on NQR	QG-06-IT-03645-2025-V2-NASSCOM
NQR Version	2

Qualification Pack

SSC/N8313: Provision, configure, test and deploy cloud infrastructure

Description

This unit is about implementing appropriate cloud architecture and provisioning, configuring, and deploying the infrastructure required for the cloud solutions.

Scope

The scope covers the following :

- Drive architecture implementation
- Provision, configure & deploy cloud infrastructure
- Ensure scalability & interoperability
- Ensure security
- Troubleshoot issues
- Implement tools & processes

Elements and Performance Criteria

Drive architecture implementation

To be competent, the user/individual on the job must be able to:

PC1. implement appropriate multi-cloud/hybrid cloud architecture and solutions

Provision, configure & deploy cloud infrastructure

To be competent, the user/individual on the job must be able to:

PC2. provision, configure, test, and deploy cloud infrastructure and related components

PC3. build, configure and deploy containers and virtual environments on the cloud

Ensure scalability & interoperability

To be competent, the user/individual on the job must be able to:

PC4. define and predict capacity requirements of the deployed virtual environment

PC5. implement multi-routing, switching, and load balancing solutions to ensure scalability of infrastructure

PC6. build automated operational processes to ensure scalable cloud deployments

PC7. ensure interoperability of cloud platform with other cloud service providers and on-premise/legacy systems

Ensure security

To be competent, the user/individual on the job must be able to:

PC8. implement cloud solutions by adhering to security standards

Troubleshoot issues

To be competent, the user/individual on the job must be able to:

PC9. troubleshoot network infrastructure in case of failure

PC10. troubleshoot issues related to the provisioning of compute and storage systems

Implement tools & processes

To be competent, the user/individual on the job must be able to:

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PC11. establish and implement services, processes, and tools to support cloud infrastructure

Cloud Service and deployment models

To be competent, the user/individual on the job must be able to:

PC12. Understand the different service models (IaaS, PaaS, SaaS)

PC13. Understand the cloud deployment models (Public, Private, and Hybrid cloud)

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** organizational policies, procedures, and guidelines that relate to provisioning, configuring, and deploying infrastructure for cloud
- KU2.** organizational policies and procedures for sharing data
- KU3.** organizational policies and procedures for documenting processes to support the deployed infrastructure
- KU4.** who to involve while configuring and deploying infrastructure on the cloud
- KU5.** the range of standard templates and tools available and how to use them
- KU6.** different types of cloud deployment schemas
- KU7.** how to define infrastructure requirements
- KU8.** how to provision infrastructure for cloud implementation
- KU9.** how to deploy and configure infrastructure on the cloud
- KU10.** different types of cloud-native and cloud-first frameworks
- KU11.** different types of load balancing techniques and solutions
- KU12.** different types of tools for provisioning and managing cloud infrastructure
- KU13.** how to test for interoperability of cloud solutions
- KU14.** different processes for supporting infrastructure deployed on cloud
- KU15.** how to automate regular and repetitive processes
- KU16.** different types of security standards and protocols
- KU17.** different types of compliance and regulatory standards
- KU18.** different types of tests to assess the performance of deployed cloud infrastructure

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** follow instructions, guidelines, procedures, rules and service level agreements
- GS2.** contribute to the quality of team working
- GS3.** communicate with others in writing
- GS4.** analyze the impact of architectural and design decisions on business and organization
- GS5.** analyze the business impact and disseminate relevant information to others
- GS6.** check your work is complete and free from errors
- GS7.** apply balanced judgments to different situations

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- GS8.** carry out rule-based transactions in line with customer-specific guidelines, procedures, rules, and service level agreements
- GS9.** check that own/peers' work meets customer requirements
- GS10.** deliver consistent and reliable service to customers
- GS11.** follow rule-based decision-making processes
- GS12.** plan and organize your own work to achieve targets and deadlines
- GS13.** apply problem-solving approaches in different situations

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Drive architecture implementation</i>	2	4	-	2
PC1. implement appropriate multi-cloud/hybrid cloud architecture and solutions	2	4	-	2
<i>Provision, configure & deploy cloud infrastructure</i>	4	7	-	3
PC2. provision, configure, test, and deploy cloud infrastructure and related components	2	4	-	2
PC3. build, configure and deploy containers and virtual environments on the cloud	2	3	-	1
<i>Ensure scalability & interoperability</i>	10	17	-	6
PC4. define and predict capacity requirements of the deployed virtual environment	2	4	-	1
PC5. implement multi-routing, switching, and load balancing solutions to ensure scalability of infrastructure	3	5	-	2
PC6. build automated operational processes to ensure scalable cloud deployments	3	5	-	2
PC7. ensure interoperability of cloud platform with other cloud service providers and on-premise/legacy systems	2	3	-	1
<i>Ensure security</i>	2	4	-	2
PC8. implement cloud solutions by adhering to security standards	2	4	-	2
<i>Troubleshoot issues</i>	5	8	-	4
PC9. troubleshoot network infrastructure in case of failure	2	4	-	2
PC10. troubleshoot issues related to the provisioning of compute and storage systems	3	4	-	2
<i>Implement tools & processes</i>	3	4	-	1

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. establish and implement services, processes, and tools to support cloud infrastructure	3	4	-	1
<i>Cloud Service and deployment models</i>	4	6	-	2
PC12. Understand the different service models (IaaS, PaaS, SaaS)	2	3	-	1
PC13. Understand the cloud deployment models (Public, Private, and Hybrid cloud)	2	3	-	1
NOS Total	30	50	-	20

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National Occupational Standards (NOS) Parameters

NOS Code	SSC/N8313
NOS Name	Provision, configure, test and deploy cloud infrastructure
Sector	IT-ITeS
Sub-Sector	Future Skills
Occupation	Cloud Computing
NSQF Level	6
Credits	5
Version	4.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

Qualification Pack

SSC/N8314: Maintain compute, network and storage systems deployed on cloud and ensure backup and recovery

Description

This unit is about managing infrastructure deployed on the cloud and ensuring system security, backup, and recovery in case of system failure

Scope

The scope covers the following :

- Manage systems configurations
- Maintain systems & their security
- Cloud Security and compliance
- Execute backup & recovery procedures
- Troubleshoot issues
- Automate systems provisioning

Elements and Performance Criteria

Manage systems configurations

To be competent, the user/individual on the job must be able to:

- PC1.** maintain configuration of compute, storage, and network systems deployed on the cloud
PC2. test new deployment configurations for bugs and errors

Maintain systems & their security

To be competent, the user/individual on the job must be able to:

- PC3.** update compute and storage systems with security patches and updates
PC4. perform network maintenance, upgrades, and refreshes after consultation with the cloud solution provider
PC5. ensure security protocols are followed across deployed systems

Cloud Security and compliance

To be competent, the user/individual on the job must be able to:

- PC6.** ensure cloud security and compliance in the systems

Execute backup & recovery procedures

To be competent, the user/individual on the job must be able to:

- PC7.** backup and maintain storage and systems configuration data
PC8. execute recovery procedures in case of storage and systems failures

Troubleshoot issues

To be competent, the user/individual on the job must be able to:

- PC9.** troubleshoot and resolve issues related to the operation and provisioning of cloud systems using appropriate diagnostic and monitoring tools

Automate systems provisioning

To be competent, the user/individual on the job must be able to:

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- PC10.** work closely with DevSecOps teams to build and support continuous integration and deployment
- PC11.** develop new processes to automate infrastructure operations
- PC12.** develop automation scripts for cloud provisioning and release management
- PC13.** leverage cloud configuration management tools to automate systems integration, deployment, and management
- PC14.** Leverage infrastructure automation tools to enable scaling up of infrastructure

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** organizational policies, procedures, and guidelines that relate to managing systems configuration, security, backup, and recovery
- KU2.** organizational policies and procedures for sharing data
- KU3.** organizational policies and procedures for documenting processes related to configuring and deploying infrastructure on the cloud
- KU4.** who to involve while configuring and deploying infrastructure on the cloud
- KU5.** the range of standard templates and tools available and how to use them
- KU6.** different types of cloud deployment schemas
- KU7.** how to provision infrastructure for cloud implementation
- KU8.** how to deploy infrastructure on the cloud
- KU9.** how to manage the configuration of the infrastructure deployed on the cloud
- KU10.** how to update and upgrade different types of cloud infrastructure
- KU11.** different types of security standards and protocols
- KU12.** how to manage user access to deployed resources
- KU13.** how to create a backup of systems configuration
- KU14.** how to create a backup of data stored on the cloud and recover lost data
- KU15.** different principles and practices of DevSecOps
- KU16.** how to automate the provisioning and management of cloud infrastructure
- KU17.** different types of tools to automate provisioning and management of cloud infrastructure
- KU18.** different types of load balancing techniques and solutions
- KU19.** different types of tools to auto-scale cloud infrastructure

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** follow instructions, guidelines, procedures, rules, and service level agreements
- GS2.** contribute to the quality of teamwork
- GS3.** communicate with others in writing
- GS4.** analyze the impact of architectural and design decisions on business and organization

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- GS5.** analyze data and activities to disseminate relevant information to others
- GS6.** check your work is complete and free from errors
- GS7.** apply balanced judgments to different situations
- GS8.** carry out rule-based transactions in line with customer-specific guidelines, procedures, rules, and service level agreements
- GS9.** check that own/peers work meets customer requirements
- GS10.** deliver consistent and reliable service to customers
- GS11.** plan and organize the work to achieve targets and meet deadlines
- GS12.** apply problem-solving approaches to different situations
- GS13.** inform system anomalies to the supervisor

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Manage systems configurations</i>	6	10	-	4
PC1. maintain configuration of compute, storage, and network systems deployed on the cloud	3	5	-	2
PC2. test new deployment configurations for bugs and errors	3	5	-	2
<i>Maintain systems & their security</i>	7	10	-	4
PC3. update compute and storage systems with security patches and updates	2	3	-	1
PC4. perform network maintenance, upgrades, and refreshes after consultation with the cloud solution provider	3	4	-	2
PC5. ensure security protocols are followed across deployed systems	2	3	-	1
<i>Cloud Security and compliance</i>	2	3	-	1
PC6. ensure cloud security and compliance in the systems	2	3	-	1
<i>Execute backup & recovery procedures</i>	5	9	-	3
PC7. backup and maintain storage and systems configuration data	3	5	-	2
PC8. execute recovery procedures in case of storage and systems failures	2	4	-	1
<i>Troubleshoot issues</i>	2	4	-	2
PC9. troubleshoot and resolve issues related to the operation and provisioning of cloud systems using appropriate diagnostic and monitoring tools	2	4	-	2
<i>Automate systems provisioning</i>	8	14	-	6
PC10. work closely with DevSecOps teams to build and support continuous integration and deployment	2	3	-	1

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. develop new processes to automate infrastructure operations	1	2	-	1
PC12. develop automation scripts for cloud provisioning and release management	1	2	-	1
PC13. leverage cloud configuration management tools to automate systems integration, deployment, and management	1	2	-	1
PC14. Leverage infrastructure automation tools to enable scaling up of infrastructure	3	5	-	2
NOS Total	30	50	-	20

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National Occupational Standards (NOS) Parameters

NOS Code	SSC/N8314
NOS Name	Maintain compute, network and storage systems deployed on cloud and ensure backup and recovery
Sector	IT-ITeS
Sub-Sector	Future Skills
Occupation	Cloud Computing
NSQF Level	6
Credits	3
Version	4.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

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SSC/N8315: Automate cloud infrastructure management

Description

This unit is about automating IT infrastructure management on the cloud to reduce manual efforts associated with provisioning, securing, and managing cloud computing workloads.

Scope

The scope covers the following :

- Manage configurations
- Automate and orchestrate processes
- Adhere to security policies

Elements and Performance Criteria

Manage configurations

To be competent, the user/individual on the job must be able to:

- PC1.** automate and maintain infrastructure configuration and operating procedures for cloud provisioning and release management

Automate and orchestrate processes

To be competent, the user/individual on the job must be able to:

- PC2.** integrate cloud automation tools with the deployed cloud systems and processes
- PC3.** leverage cloud configuration management tools and orchestration tools to automate systems deployment and management
- PC4.** implement, manage, and automate load balancing tools to scale up infrastructure on the cloud as and when required
- PC5.** leverage tools to automate the container creation process
- PC6.** leverage container orchestration tools to automate container deployment and management

Adhere to security policies

To be competent, the user/individual on the job must be able to:

- PC7.** ensure that automated processes are secure and adhere to the security policies of the organization

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** organizational policies, procedures, and guidelines that relate to the provisioning of systems on the cloud and managing systems configuration
- KU2.** organizational policies and procedures for sharing data
- KU3.** organizational policies and procedures for documenting processes related to configuring and deploying infrastructure on the cloud
- KU4.** who to involve while automating cloud infrastructure management

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- KU5.** the range of standard templates and tools available and how to use them
- KU6.** how to provision infrastructure for cloud implementation
- KU7.** how to deploy infrastructure on the cloud
- KU8.** how to manage the configuration of the infrastructure deployed on the cloud
- KU9.** how to automate configuration of infrastructure deployed on cloud
- KU10.** how to automate standard operating procedures for provisioning and deployment of cloud infrastructure
- KU11.** different types of tools to manage and backup cloud infrastructure configuration
- KU12.** different types of tools to automate the provisioning of infrastructure on cloud
- KU13.** different types of cloud orchestration tools
- KU14.** different types of load balancing tools
- KU15.** different types of container technologies
- KU16.** how to create and deploy containers
- KU17.** different types of tools to automate the creation and deployment of containers
- KU18.** different types of security standards and protocols

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** follow instructions, guidelines, procedures, rules, and service level agreements
- GS2.** contribute to the quality of teamwork
- GS3.** work independently and collaboratively
- GS4.** pass on relevant information to others
- GS5.** check the work is complete and free from errors
- GS6.** apply balanced judgments to different situations
- GS7.** check that own/peers' work meets customer requirements
- GS8.** deliver consistent and reliable service to customers
- GS9.** plan and organize the work to achieve targets and meet deadlines
- GS10.** apply problem-solving approaches in different situations

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Manage configurations</i>	4	9	-	3
PC1. automate and maintain infrastructure configuration and operating procedures for cloud provisioning and release management	4	9	-	3
<i>Automate and orchestrate processes</i>	22	35	-	15
PC2. integrate cloud automation tools with the deployed cloud systems and processes	5	8	-	3
PC3. leverage cloud configuration management tools and orchestration tools to automate systems deployment and management	5	8	-	3
PC4. implement, manage, and automate load balancing tools to scale up infrastructure on the cloud as and when required	3	5	-	3
PC5. leverage tools to automate the container creation process	5	8	-	3
PC6. leverage container orchestration tools to automate container deployment and management	4	6	-	3
<i>Adhere to security policies</i>	4	6	-	2
PC7. ensure that automated processes are secure and adhere to the security policies of the organization	4	6	-	2
NOS Total	30	50	-	20

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National Occupational Standards (NOS) Parameters

NOS Code	SSC/N8315
NOS Name	Automate cloud infrastructure management
Sector	IT-ITeS
Sub-Sector	Future Skills
Occupation	Cloud Computing
NSQF Level	6
Credits	3
Version	4.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

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SSC/N8316: Migrate legacy/ on-premise/cloud applications and data to new cloud platforms

Description

This unit is about migrating legacy/on-premise/cloud applications and data to new cloud platforms and ensuring the migrated workload is secure and performs as intended.

Scope

The scope covers the following :

- Develop migration plan
- Drive resource provisioning
- Drive cloud migration
- Follow SOPs
- Validate migration
- Ensure alignment with governance policies
- Ensure security

Elements and Performance Criteria

Develop migration plan

To be competent, the user/individual on the job must be able to:

- PC1.** support the identification and estimation of application remediation and changes in the source code required for the preparation of migration
- PC2.** develop plans to create appropriate migration bundles and sequences and prepare data for interfaces
- PC3.** create, validate and execute fallback/ rollback plans in case the migration does not perform as intended

Drive resource provisioning

To be competent, the user/individual on the job must be able to:

- PC4.** provision storage disks, virtual storage devices, compute systems, and all other resources on the cloud
- PC5.** configure servers including storage & network for migration

Drive cloud migration

To be competent, the user/individual on the job must be able to:

- PC6.** migrate servers, storage, and databases to the new environment
- PC7.** manage the migration of traffic
- PC8.** identify and leverage appropriate migration tools to execute migration procedures

Follow SOPs

To be competent, the user/individual on the job must be able to:

- PC9.** follow and maintain Standard Operating Procedures (SOPs) for the execution of migration

Validate migration

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To be competent, the user/individual on the job must be able to:

PC10. ensure data integrity of the database and other storage information and identify gaps in desired data

PC11. implement checks and balances and identify protocols to validate successful migration

Ensure alignment with governance policies

To be competent, the user/individual on the job must be able to:

PC12. work closely with developers and database admins to align to governance policies and oversee their implementation

Ensure security

To be competent, the user/individual on the job must be able to:

PC13. evaluate proofs-of-concept for server, storage, network, and security technologies to support the cloud infrastructure

PC14. ensure migrated data adhere to security standards and guidelines as per organizational and regulatory requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. organizational policies, procedures, and guidelines that relate to the migration of applications and data to the cloud platforms

KU2. organizational policies and procedures for sharing data

KU3. organizational policies and procedures for documenting processes related to the migration of applications and data to the cloud platforms

KU4. who to involve while planning migration of applications and data to the cloud platforms

KU5. the range of standard templates and tools available and how to use them

KU6. how to evaluate changes required on the application source code before migration

KU7. how to define migration bundles and their sequence

KU8. different types of cloud migration frameworks

KU9. different methodologies for migrating workloads to the cloud

KU10. how to create a rollback plan in case of migration failure

KU11. how to deploy infrastructure on the cloud

KU12. different types of cloud migration tools

KU13. Standard Operating Procedures (SoP) for executing cloud migration

KU14. different types of security standards and policies

KU15. how to validate successful migration to the cloud

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. follow instructions, guidelines, procedures, rules, and service level agreements

GS2. contribute to the quality of teamwork

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- GS3.** work independently and collaboratively
- GS4.** analyze the business impact and disseminate relevant information to others
- GS5.** check the work is complete and free from errors
- GS6.** apply balanced judgments to different situations
- GS7.** carry out rule-based transactions in line with customer-specific guidelines, procedures, rules, and service level agreements
- GS8.** check that own/peers' work meets customer requirements
- GS9.** deliver consistent and reliable service to customers
- GS10.** work effectively in a customer-facing environment
- GS11.** plan and organize the work to achieve targets and meet deadlines
- GS12.** apply problem-solving approaches in different situations
- GS13.** inform system anomalies to the supervisor

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Develop migration plan</i>	6	12	-	6
PC1. support the identification and estimation of application remediation and changes in the source code required for the preparation of migration	2	4	-	2
PC2. develop plans to create appropriate migration bundles and sequences and prepare data for interfaces	2	4	-	2
PC3. create, validate and execute fallback/rollback plans in case the migration does not perform as intended	2	4	-	2
<i>Drive resource provisioning</i>	4	6	-	2
PC4. provision storage disks, virtual storage devices, compute systems, and all other resources on the cloud	2	3	-	1
PC5. configure servers including storage & network for migration	2	3	-	1
<i>Drive cloud migration</i>	6	11	-	5
PC6. migrate servers, storage, and databases to the new environment	2	3	-	1
PC7. manage the migration of traffic	2	4	-	2
PC8. identify and leverage appropriate migration tools to execute migration procedures	2	4	-	2
<i>Follow SOPs</i>	2	3	-	1
PC9. follow and maintain Standard Operating Procedures (SOPs) for the execution of migration	2	3	-	1
<i>Validate migration</i>	5	7	-	2
PC10. ensure data integrity of the database and other storage information and identify gaps in desired data	2	3	-	1

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. implement checks and balances and identify protocols to validate successful migration	3	4	-	1
<i>Ensure alignment with governance policies</i>	2	3	-	1
PC12. work closely with developers and database admins to align to governance policies and oversee their implementation	2	3	-	1
<i>Ensure security</i>	5	8	-	3
PC13. evaluate proofs-of-concept for server, storage, network, and security technologies to support the cloud infrastructure	1	3	-	1
PC14. ensure migrated data adhere to security standards and guidelines as per organizational and regulatory requirements	4	5	-	2
NOS Total	30	50	-	20

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National Occupational Standards (NOS) Parameters

NOS Code	SSC/N8316
NOS Name	Migrate legacy/ on-premise/cloud applications and data to new cloud platforms
Sector	IT-ITeS
Sub-Sector	Future Skills
Occupation	Cloud Computing
NSQF Level	6
Credits	3
Version	4.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

Qualification Pack

SSC/N8317: Integrate cloud infrastructure with existing IT systems

Description

This unit is about creating suitable integrations to connect cloud infrastructure with existing IT systems in the organization.

Scope

The scope covers the following :

- Gather requirements
- Plan integration activities
- Map process flows
- Integrate API/data/applications
- Perform unit testing
- Perform system testing
- Ensure system security
- Cloud costing

Elements and Performance Criteria

Gather requirements

To be competent, the user/individual on the job must be able to:

PC1. define and document integration needs, users, roles and permissions

Plan integration activities

To be competent, the user/individual on the job must be able to:

PC2. create a plan to divide the integration activity into different phases

Map process flows

To be competent, the user/individual on the job must be able to:

PC3. map the process flow and the transaction of data in between subsystems

PC4. plan to block erroneous data from being entered that disrupts the chronological events of a process

Integrate API/data/applications

To be competent, the user/individual on the job must be able to:

PC5. map upstream and downstream application integration points

PC6. plan and execute data integration elements

PC7. research on off-the-shelf APIs (Application Programming Interfaces) that supports the integration of sub-systems

PC8. build APIs to integrate sub-systems

Perform unit testing

To be competent, the user/individual on the job must be able to:

PC9. carry out unit tests to ensure desired performance from the APIs

Perform system testing

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To be competent, the user/individual on the job must be able to:

PC10. perform systems tests to verify that the requirements are met

Ensure system security

To be competent, the user/individual on the job must be able to:

PC11. ensure the systems are secure before, during, and after integration

Cloud costing

To be competent, the user/individual on the job must be able to:

PC12. Ensure the right cloud costing strategy is in place for providing a unified view of expenses across different platforms

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** organizational policies, procedures, and guidelines that relate to mapping process flows and integrating systems and sub-systems
- KU2.** organizational policies and procedures for sharing data
- KU3.** organizational policies and procedures for documenting the plan for integrating systems and sub-systems
- KU4.** who to involve while mapping process flows and creating the integration plan
- KU5.** the range of standard templates and tools available and how to use them
- KU6.** how to define process flows
- KU7.** how to define integration requirements
- KU8.** how to create an integration plan
- KU9.** different methods of systems integration
- KU10.** how to define upstream and downstream software dependencies
- KU11.** how to build APIs for systems integration
- KU12.** different off-the-shelf APIs available for systems integration
- KU13.** different types of systems, integration, and security tests
- KU14.** how to carry out unit tests
- KU15.** different types of security standards and protocols

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** ask for clarification and advice from appropriate people
- GS2.** follow instructions, guidelines, procedures, rules, and service level agreements
- GS3.** contribute to the quality of teamwork
- GS4.** analyze the impact of architectural and design decisions on business and organization
- GS5.** analyze the business impact and disseminate relevant information to others
- GS6.** check the work is complete and free from errors

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- GS7.** apply balanced judgments to different situations
- GS8.** check that own/peers' work meets customer requirements
- GS9.** deliver consistent and reliable service to customers
- GS10.** work effectively in a customer-facing environment
- GS11.** plan and organize the work to achieve targets and meet deadlines
- GS12.** apply problem-solving approaches to different situations

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Gather requirements</i>	3	4	-	1
PC1. define and document integration needs, users, roles and permissions	3	4	-	1
<i>Plan integration activities</i>	3	4	-	2
PC2. create a plan to divide the integration activity into different phases	3	4	-	2
<i>Map process flows</i>	6	8	-	4
PC3. map the process flow and the transaction of data in between subsystems	3	4	-	2
PC4. plan to block erroneous data from being entered that disrupts the chronological events of a process	3	4	-	2
<i>Integrate API/data/applications</i>	9	16	-	7
PC5. map upstream and downstream application integration points	3	4	-	2
PC6. plan and execute data integration elements	2	4	-	2
PC7. research on off-the-shelf APIs (Application Programming Interfaces) that supports the integration of sub-systems	2	4	-	2
PC8. build APIs to integrate sub-systems	2	4	-	1
<i>Perform unit testing</i>	2	4	-	1
PC9. carry out unit tests to ensure desired performance from the APIs	2	4	-	1
<i>Perform system testing</i>	3	5	-	2
PC10. perform systems tests to verify that the requirements are met	3	5	-	2
<i>Ensure system security</i>	2	5	-	2

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. ensure the systems are secure before, during, and after integration	2	5	-	2
<i>Cloud costing</i>	2	4	-	1
PC12. Ensure the right cloud costing strategy is in place for providing a unified view of expenses across different platforms	2	4	-	1
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SSC/N8317
NOS Name	Integrate cloud infrastructure with existing IT systems
Sector	IT-ITeS
Sub-Sector	Future Skills
Occupation	Cloud Computing
NSQF Level	6
Credits	4
Version	4.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

Qualification Pack

DGT/VSQ/N0102: Employability Skills (60 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** identify employability skills required for jobs in various industries
- PC2.** identify and explore learning and employability portals

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

- PC3.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC4.** follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC5.** recognize the significance of 21st Century Skills for employment
- PC6.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

Basic English Skills

To be competent, the user/individual on the job must be able to:

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- PC7.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC8.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC9.** write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC10.** understand the difference between job and career
- PC11.** prepare a career development plan with short- and long-term goals, based on aptitude

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC12.** follow verbal and non-verbal communication etiquette and active listening techniques in various settings
- PC13.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC14.** communicate and behave appropriately with all genders and PwD
- PC15.** escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC16.** select financial institutions, products and services as per requirement
- PC17.** carry out offline and online financial transactions, safely and securely
- PC18.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC19.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

- PC20.** operate digital devices and carry out basic internet operations securely and safely
- PC21.** use e- mail and social media platforms and virtual collaboration tools to work effectively
- PC22.** use basic features of word processor, spreadsheets, and presentations

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC23.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC24.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC25.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC26.** identify different types of customers
- PC27.** identify and respond to customer requests and needs in a professional manner.

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PC28. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC29. create a professional Curriculum vitae (Résumé)

PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC31. apply to identified job openings using offline /online methods as per requirement

PC32. answer questions politely, with clarity and confidence, during recruitment and selection

PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills and different learning and employability related portals

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

KU6. importance of career development and setting long- and short-term goals

KU7. about effective communication

KU8. POSH Act

KU9. Gender sensitivity and inclusivity

KU10. different types of financial institutes, products, and services

KU11. how to compute income and expenditure

KU12. importance of maintaining safety and security in offline and online financial transactions

KU13. different legal rights and laws

KU14. different types of digital devices and the procedure to operate them safely and securely

KU15. how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

KU16. how to identify business opportunities

KU17. types and needs of customers

KU18. how to apply for a job and prepare for an interview

KU19. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and write different types of documents/instructions/correspondence

GS2. communicate effectively using appropriate language in formal and informal settings

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- GS3.** behave politely and appropriately with all
- GS4.** how to work in a virtual mode
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. identify employability skills required for jobs in various industries	-	-	-	-
PC2. identify and explore learning and employability portals	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC4. follow environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	2	4	-	-
PC5. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC7. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-
PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC9. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. understand the difference between job and career	-	-	-	-
PC11. prepare a career development plan with short- and long-term goals, based on aptitude	-	-	-	-
<i>Communication Skills</i>	2	2	-	-
PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings	-	-	-	-
PC13. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	2	-	-
PC14. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC15. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC16. select financial institutions, products and services as per requirement	-	-	-	-
PC17. carry out offline and online financial transactions, safely and securely	-	-	-	-
PC18. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	3	4	-	-
PC20. operate digital devices and carry out basic internet operations securely and safely	-	-	-	-
PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively	-	-	-	-
PC22. use basic features of word processor, spreadsheets, and presentations	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Entrepreneurship</i>	2	3	-	-
PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC26. identify different types of customers	-	-	-	-
PC27. identify and respond to customer requests and needs in a professional manner.	-	-	-	-
PC28. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	3	-	-
PC29. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC31. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC32. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0102
NOS Name	Employability Skills (60 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	12/03/2026
Next Review Date	12/03/2029
NSQC Clearance Date	12/03/2026

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

The assessment will consist of a blend of hands-on practical evaluations, viva-voce, and online proctored scenario-based multiple-choice questions ensuring a thorough evaluation of the individual's proficiency in learning outcomes, practical understanding, and real-world application of concepts.

Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

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National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
SSC/N8313.Provision, configure, test and deploy cloud infrastructure	30	50	-	20	100	18
SSC/N8314.Maintain compute, network and storage systems deployed on cloud and ensure backup and recovery	30	50	-	20	100	18
SSC/N8315.Automate cloud infrastructure management	30	50	-	20	100	18
SSC/N8316.Migrate legacy/ on-premise/cloud applications and data to new cloud platforms	30	50	-	20	100	18
SSC/N8317.Integrate cloud infrastructure with existing IT systems	30	50	-	20	100	18
DGT/VSQ/N0102.Employability Skills (60 Hours)	20	30	-	-	50	10
Total	170	280	-	100	550	100

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

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Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.